702 PS-5

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Problem Set - 3

3. Problem to demonstrate the role of qualitative (ordinal) predictors in addition to quantitative predictors in multiple linear regression

Consider “diamonds” data set in R. It is in the ggplot2 package. Make a list of all the ordinal categorical variables. Identify the response.

(a) Run a linear regression of the response on the quality of cut. Write the fitted regression model.

```
library(ggplot2)
str(diamonds)
```

```
## tibble [53,940 × 10] (S3: tbl_df/tbl/data.frame)
##  $ carat  : num [1:53940] 0.23 0.21 0.23 0.29 0.31 0.24 0.24 0.26 0.22
0.23 ...
##  $ cut    : Ord.factor w/ 5 levels "Fair"<"Good"<..: 5 4 2 4 2 3 3 3 1 3
...
##  $ color  : Ord.factor w/ 7 levels "D"<"E"<"F"<"G"<..: 2 2 2 6 7 7 6 5 2 5
...
##  $ clarity: Ord.factor w/ 8 levels "I1"<"SI2"<"SI1"<..: 2 3 5 4 2 6 7 3 4
5 ...
##  $ depth  : num [1:53940] 61.5 59.8 56.9 62.4 63.3 62.8 62.3 61.9 65.1
59.4 ...
##  $ table  : num [1:53940] 55 61 65 58 58 57 57 55 61 61 ...
##  $ price  : int [1:53940] 326 326 327 334 335 336 336 337 337 338 ...
##  $ x      : num [1:53940] 3.95 3.89 4.05 4.2 4.34 3.94 3.95 4.07 3.87 4
...
##  $ y      : num [1:53940] 3.98 3.84 4.07 4.23 4.35 3.96 3.98 4.11 3.78
4.05 ...
##  $ z      : num [1:53940] 2.43 2.31 2.31 2.63 2.75 2.48 2.47 2.53 2.49
2.39 ...
```

```
grade=as.numeric(diamonds$cut)
f=function(x){
  if(x-5==0){return(c(0,0,0,0))
  }else if(x-5==-1){return(c(1,0,0,0))
  }else if(x-5==-2){return(c(1,1,0,0))
  }else if(x-5==-3){return(c(1,1,1,0))
  }else{return(c(1,1,1,1))}}
p=t(sapply(grade,f))
colnames(p)=c("Premium","VGood","Good","Fair")
```

```r
diamonds=cbind(diamonds,p)
summary(lm(price~Premium+VGood+Good+Fair,data=diamonds))

##
## Call:
## lm(formula = price ~ Premium + VGood + Good + Fair, data = diamonds)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -4258  -2741  -1494   1360  15348
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3457.54      27.00 128.051  < 2e-16 ***
## Premium      1126.72      43.22  26.067  < 2e-16 ***
## VGood        -602.50      49.39 -12.198  < 2e-16 ***
## Good          -52.90      67.10  -0.788  0.43055
## Fair          429.89     113.85   3.776  0.00016 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3964 on 53935 degrees of freedom
## Multiple R-squared:  0.01286,    Adjusted R-squared:  0.01279
## F-statistic: 175.7 on 4 and 53935 DF,  p-value: < 2.2e-16
```

(b) Test whether the expected price of diamond with premium cut is significantly different from that of the ideal cut.

We will perform the following test:

$H_0: \beta_{\text{Premium}} - \beta_{\text{Ideal}} = 0$ ***ag*** $H_1: \beta_{\text{Premium}} - \beta_{\text{Ideal}} \neq 0$

The test statistic under H_0 is given as:

$$t = \frac{\hat{\beta}_{\text{Premium}}}{SE(\hat{\beta}_{\text{Premium}})}$$

The p-value we observe for the test is negligible (almost equal to 0 << $\alpha = 0.05$). Therefore, we conclude that there is a statistically significant difference between the two levels.

(c) What is the expected price of a diamond of ideal cut?

The expected price of a diamond of ideal cut is $3457.54.

(d) Modify the regression model in (a) by incorporating the predictor "table". Write the fitted regression model.

```r
summary(lm(price~Premium+VGood+Good+Fair+table,data=diamonds))

##
## Call:
## lm(formula = price ~ Premium + VGood + Good + Fair + table, data =
```

```
diamonds)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -5630  -2694  -1458   1346  15690
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6563.672    517.450 -12.685  < 2e-16 ***
## Premium       626.220     50.215  12.471  < 2e-16 ***
## VGood        -461.015     49.761  -9.265  < 2e-16 ***
## Good         -185.162     67.220  -2.755  0.00588 **
## Fair          365.568    113.504   3.221  0.00128 **
## table         179.105      9.236  19.393  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3950 on 53934 degrees of freedom
## Multiple R-squared:  0.0197, Adjusted R-squared:  0.01961
## F-statistic: 216.7 on 5 and 53934 DF,  p-value: < 2.2e-16
```

(e) Test for the significance of "table" in predicting the price of diamond.

For this we will be performing the student's t test.

We will perform the following test:

$H_0: \beta_{table} = 0$ ***ag*** $H_1: \beta_{table} \neq 0$

The test statistic under H_0 is given as:

t = $\frac{\widehat{\beta}_{table}}{SE(\beta_{table})}$

The p-value we observe for the test is negligible (almost equal to 0 << $\alpha = 0.05$). Therefore, we conclude that table is a statistically significant predictor.

(f) Find the average estimated price of a diamond with an average table value and which is of fair cut.

The average estimated price of a diamond with an average table value and which is of fair cut is $4092.76

Problem Set - 5

1. Problem to demonstrate the utility of K nearest neighbour regression over least squares regression.

Consider a setting with n = 1000 observations. Generate:

(i) x_{1i} from $N(0, 2^2)$ and x_{2i} from $Poisson(\lambda = 1.5)$.

(ii) ε_i from $N(0,1)$.

(iii) $yi = -2 + 1.4x_{1i} - 2.6x_{2i} + \varepsilon_i$

We are using the seed as 123.

Split the data into train and test sets. Keep the first 800 observations as training data and the remaining as test data. Work out the following:

```
set.seed(123)
n=1000
x1=rnorm(n,0,2);x2=rpois(n,1.5);e=rnorm(n)
y=-2+1.4*x1-2.6*x2+e
d=data.frame(y,x1,x2)
head(d)

##            y         x1 x2
## 1 -4.3903185 -1.1209513  0
## 2 -2.9517542 -0.4603550  0
## 3  1.4622853  3.1174166  0
## 4 -3.7755078  0.1410168  1
## 5 -3.1176393  0.2585755  1
## 6 -0.2706045  3.4301300  2

gg=c(1:800)
train=d[gg,];test=d[-gg,]
```

1. Fit a multiple linear regression equation of y on x1 and x2. Calculate test MSE.

```
m1=lm(y~x1+x2,data=train)
summary(m1)

##
## Call:
## lm(formula = y ~ x1 + x2, data = train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.0727 -0.6573 -0.0125  0.6921  3.2412
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.07300    0.05382  -38.52   <2e-16 ***
## x1           1.38207    0.01767   78.21   <2e-16 ***
## x2          -2.55584    0.02768  -92.34   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.98 on 797 degrees of freedom
```

```
## Multiple R-squared:  0.9492, Adjusted R-squared:  0.9491
## F-statistic:  7445 on 2 and 797 DF,  p-value: < 2.2e-16

yhat=predict(m1,newdata=test)
test.mse=mean((yhat-test[,1])^2)
test.mse

## [1] 0.998901
```

2. Fit a KNN model with k = 1, 2, 5, 9, 15. Calculate test MSE for each choice of k.

```
library(ggplot2)
library(lattice)
library(caret)
mseks=c()
ks=c(1,2,5,9,15)
for(i in 1:5)
{kreg=knnreg(y~x1+x2,data=train,k=ks[i])
 ykreg=predict(kreg,test)
 mseks[i]=mean((test$y-ykreg)^2)}
output=data.frame(ks,mseks,rep(test.mse,each=5))
colnames(output)=c("Choices of k","Test MSE under KNN","Test MSE under
Multiple Linear Reg")
output

##   Choices of k Test MSE under KNN Test MSE under Multiple Linear Reg
## 1            1           2.219793                           0.998901
## 2            2           1.729587                           0.998901
## 3            5           1.303978                           0.998901
## 4            9           1.205371                           0.998901
## 5           15           1.232730                           0.998901
```

Now we will be considering the function:

$$y_i = \frac{1}{(-2 + 1.4x_{1i} - 2.6x_{2i} + 2.9x_{1i}^2)} + 3.1sinx_{2i} - 1.5x_{1i}x_{2i}^2 + \varepsilon_i$$

Work out the problems in (1) and (2). Compare and comment on the results.

```
y1=1/(-2+1.4*x1-2.6*x2+2.9*(x1^2))+3.1*sin(x2)-1.5*x1*(x2^2)+e
d1=data.frame(y1,x1,x2)
head(d1)

##           y1         x1 x2
## 1  12.581968 -1.1209513  0
## 2  -0.799890 -0.4603550  0
## 3  -0.869362  3.1174166  0
## 4   2.793949  0.1410168  1
## 5   3.093778  0.2585755  1
## 6 -15.603221  3.4301300  2
```

```r
train1=d1[gg,];test1=d1[-gg,]
m2=lm(y1~x1+x2,data=train1)
summary(m2)
```

```
##
## Call:
## lm(formula = y1 ~ x1 + x2, data = train1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -241.819   -5.150   -0.051    5.566  155.662
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.2369     1.0272  -0.231  0.81764
## x1           -6.3747     0.3373 -18.901  < 2e-16 ***
## x2            1.3849     0.5283   2.621  0.00892 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.7 on 797 degrees of freedom
## Multiple R-squared:  0.3146, Adjusted R-squared:  0.3129
## F-statistic: 182.9 on 2 and 797 DF,  p-value: < 2.2e-16
```

```r
y1test=predict(m2,newdata=test1)
testmse1=mean((y1test-test1[,1])^2);testmse1
```

```
## [1] 205.1776
```

```r
msek1=c()
for(i in 1:5)
{kr1=knnreg(y1~x1+x2,data=train1,k=ks[i])
 y1_knn=predict(kr1,test1)
 msek1[i]=mean((test1[,1]-y1_knn)^2)}
output2=data.frame(ks,msek1,rep(testmse1,each=5))
colnames(output2)=c("Choices of k","Test MSE under KNN","Test MSE under
Multiple Linear Reg")
output2
```

```
##   Choices of k Test MSE under KNN Test MSE under Multiple Linear Reg
## 1            1           47.52490                           205.1776
## 2            2           54.48942                           205.1776
## 3            5           59.77963                           205.1776
## 4            9           62.10913                           205.1776
## 5           15           63.72303                           205.1776
```

From the above comparison, we observe that the KNN method performs significantly better than multiple linear regression, as it produces a much lower test mean squared error. This is because KNN can capture non-linear relationships, whereas multiple linear regression assumes linearity and therefore performs poorly in this case. Although the minimum MSE occurs when

k=1, the increase in MSE for larger values of k is relatively small, suggesting that choosing a moderately large value of k would still be appropriate.